Compressor Rotor Joints and Interference Fit Study

Critical Design Review

By:
Jireh Velasquez
Isaiah Guerrero
Aubrey Perez
Isabela Jimenez

Senior Project
Sponsor: Manuel Aldana (Solar Turbines)
California Polytechnic State
University, San Luis Obispo
College of Engineering

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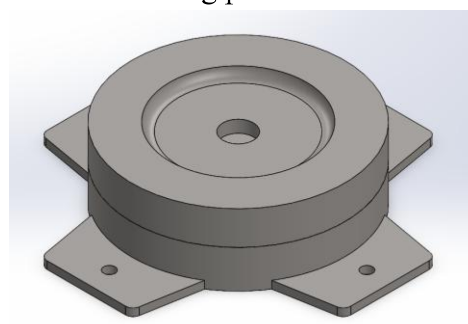
Project Overview	Concept Description
Solar Turbines needs a comprehensive study tool on the effects radial interference fits have on turbine compressor rotor stiffness. The weldments of the compressor have press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis as well as practically tested through modal impacts on manufactured prototypes to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model tool to eventually be presented to Solar Turbines.	Our project will include the creation of an Ansys model of Solar Turbine's rotor disks and three manufactured assemblies that will be tested to validate the Ansys model. These include two assemblies with dowel pin connections and one assembly welded together. Each will be mounted to a shake table and struck with an impact hammer to produce vibrations to be read with accelerometers. Each assembly is estimated to cost around $120, with the majority of the funds being used on specialized tooling for the manufacturing process. 
Design Specifications	**What's Next**
The specifications for this design include the accurate modeling of radial interference effects on rotor stiffness from the pilot fit and dowel pins, along with qualitative verification from testing. The modeling and testing of only radial interferences was decided between Solar Turbines and the team due to complications that would arise with manufacturing axial interference in the physical testing model. Other simplifications made to the design for the models, such as dowel pin size and lack of contours, allow for feasible manufacturing, while also meeting the requirements on requested radial interference qualitative data.	The next steps include finalizing results from the Ansys model parameterization of multiple configurations, manufacturing and testing of the three physical verification configurations, and a final design review presentation to Solar Turbines to provide analysis results. The Ansys model results will be recorded by the end of the winter quarter. The first configuration of the verification testing will be manufactured also by the end of the winter quarter. During spring quarter, the remaining verification testing models will be created, and all three models will be tested in the vibrations laboratory. Finally, the final presentation of this project's results will be given to Solar Turbines in-person at San Diego.

Figure 1. Four-Panel Chart for Critical Design Review.

Overview

The goal of this project is to complete a comprehensive study on how the radial interference fit of the mating dowel pins and pilot joints affect the rotor stiffness on Solar Turbine's Titan 130 turbine compressor. An Ansys model has been developed, where dowel pin and pilot fit interference levels have been parameterized to find trends in the effected rotor stiffness via natural frequency. A study on pilot fit length has also started with the Ansys model, finding the stiffness of the assembly when the length of the pilot fit is increased. Three assemblies will be manufactured to validate the results of the Ansys model. Although the Ansys model will be used to find the best theorized configurations, two configurations that will be the easiest to manufacture will be chosen for two physical assemblies. In this way, the Ansys model can still be verified while keeping the manufacturing feasible given the resources at Cal Poly. The third physical assembly will be a rigid welded one to act as a datum for the qualitative data of the two other assemblies.

The goal of this report is to explain in detail the progress of the Compressor Rotor Joints and Fits Interference Study. This report will first contain our Concept Description, which will go into further detail on our Ansys model and respective manufactured verification prototypes. Next will be our Design Justification, which will explain the decisions made regarding the required design specifications. Then, the Manufacturing Plan section will describe the detailed procedure for manufacturing our verification prototypes. Next, the Design Verification section will talk about the testing plan for the assemblies. Finally, the Budget section will break down the cost analysis performed so far.

Concept Description

The objective of this project is to quantify how the stiffness of a rotor plate assembly changes as a function of interference fit parameters. Solar Turbines is interested in understanding how variations in dowel pin interference, pilot fit interference, and pilot contact length influence the global stiffness of a gas generator rotor plate interface. Assembly stiffness directly affects natural frequency, which is critical for vibration performance and avoidance of resonance in turbine systems. A clear understanding of this relationship allows for improved manufacturing and design decisions that enhance structural reliability and dynamic stability.

The theoretical foundation for this work comes from vibration analysis of a single degree of freedom system. The governing equation of motion shows that natural frequency is proportional to the square root of stiffness divided by mass. Rearranging this relationship shows that stiffness is equal to mass multiplied by the square of the natural frequency. This equation forms the basis of the methodology used in this study. By extracting natural frequencies from a preloaded finite element model and multiplying the squared frequency by the total mass of the assembly, the effective stiffness of the assembly can be determined. This approach directly links measurable vibratory behavior to structural stiffness generated by interference fits.

The final design consists of a scaled rotor plate assembly that represents the interface between turbine rotor components. The assembly includes two circular plates joined by a central pilot interface and constrained radially by dowel pins. Controlled interference fits are introduced at both the dowel pin interfaces and the pilot fit interface. When assembled, these interferences generate contact pressure between mating surfaces. The resulting compressive stresses increase resistance to relative motion, thereby increasing the effective stiffness of the structure. As stiffness increases, natural frequency correspondingly increases according to vibration theory. The design therefore enables controlled manipulation of preload to observe changes in structural stiffness.

Full-scale Solar Turbines rotor plates are approximately twenty inches in diameter and exceed the dimensional and weight limitations of the Cal Poly vibrations laboratory equipment. The laboratory shake table has size constraints and a maximum load capacity of one hundred pounds. For this reason, a geometrically scaled assembly with a diameter of seven inches and an approximate total weight of twenty pounds was developed. Although reduced in size, the scaled assembly preserves the essential contact mechanics and geometric relationships necessary to observe stiffness trends. The simplified design allows reliable laboratory testing while maintaining the physical mechanisms that govern stiffness generation in the full-scale assembly. Axial tolerances were removed from the study to isolate the effects of radial interference. By eliminating unnecessary variables, the experiment focuses strictly on how radial preload influences assembly stiffness.

Finite element modeling was performed in Ansys Workbench to replicate the experimental configuration as accurately as possible. The geometry was constructed directly within Ansys DesignModeler to take advantage of full parameterization capabilities. Parameterization allows geometric dimensions such as dowel hole diameter, pilot bore diameter, interference magnitude, and pilot contact length to be defined as adjustable variables. These variables can then be systematically modified through a design point table. When a new design point is executed, Ansys automatically updates the geometry, resolves the static structural contact analysis, transfers the resulting stress state to a modal analysis, and outputs the updated natural frequencies. This automated workflow ensures consistency between iterations and significantly reduces manual modeling time.

The analysis sequence begins with a static structural contact simulation. Frictional contact conditions are applied at all interfaces where interference occurs. Large deformation effects are enabled to account for nonlinear contact behavior that arises from interference induced preload. A fixed boundary condition is applied to the bottom surface of one plate to replicate the experimental mounting condition used during testing. A refined tetrahedral mesh is generated with increased element density in contact regions to capture stress gradients accurately. Mesh refinement is continued until the maximum allowable node count for the student license is reached. The static solution captures the preload stresses generated by interference fits and establishes the initial stress state of the assembly.

Following completion of the static analysis, a modal analysis is performed using the preloaded stress condition. This preloaded modal analysis is essential because it incorporates the stiffness contribution from contact pressure. Without preload, the modal solution would neglect the additional stiffness generated by interference fits. The first several natural frequencies are extracted, and the dominant bending mode is selected for stiffness calculation. The stiffness of each configuration is computed by multiplying the assembly mass by the square of the extracted natural frequency. Because mass remains constant between configurations, changes in natural frequency directly reflect changes in stiffness.

Interference values were selected based on ANSI B4.1 standards for locational interference fits. Locational fits are appropriate when positional accuracy between components is critical, which is the case for turbine rotor plates. The standard defines several classes of interference magnitude, including light, medium, and heavy drive fits. Solar Turbines currently operates within the lighter interference classes for dowel pins. Therefore, the parametric study was conducted across a range spanning from light to moderate interference values. Fits exceeding this range were not considered due to manufacturing difficulty and reduced practicality. A similar approach was applied to the pilot fit. The pilot contact length was treated as a separate variable with an upper limit defined as one half of the total plate thickness. This constraint ensured that the design remained realistic and manufacturable.

Results from the parametric study revealed that increasing dowel interference alone produced only minor increases in stiffness.

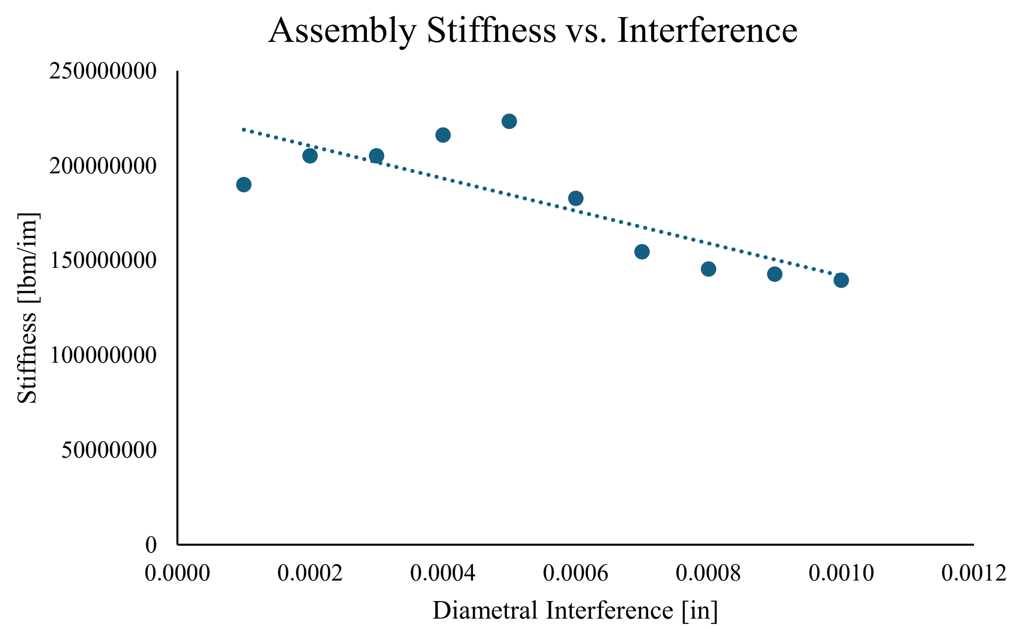


Figure 1: Extracting the parameterized results and plotting them in excel reveals unexpected stiffness results.

Increasing pilot diametral interference similarly resulted in modest stiffness gains.

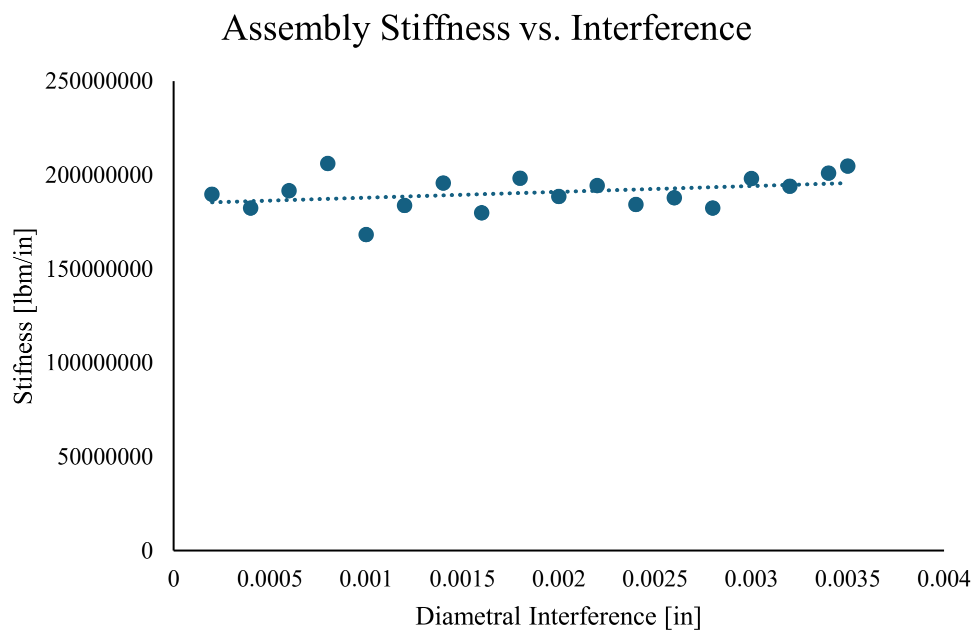


Figure 2: Iterating interferences between the pilot fit began to show more of an increase in assembly stiffness as diametral interference increased.

In contrast, increasing pilot contact length produced a significant increase in natural frequency and therefore stiffness.

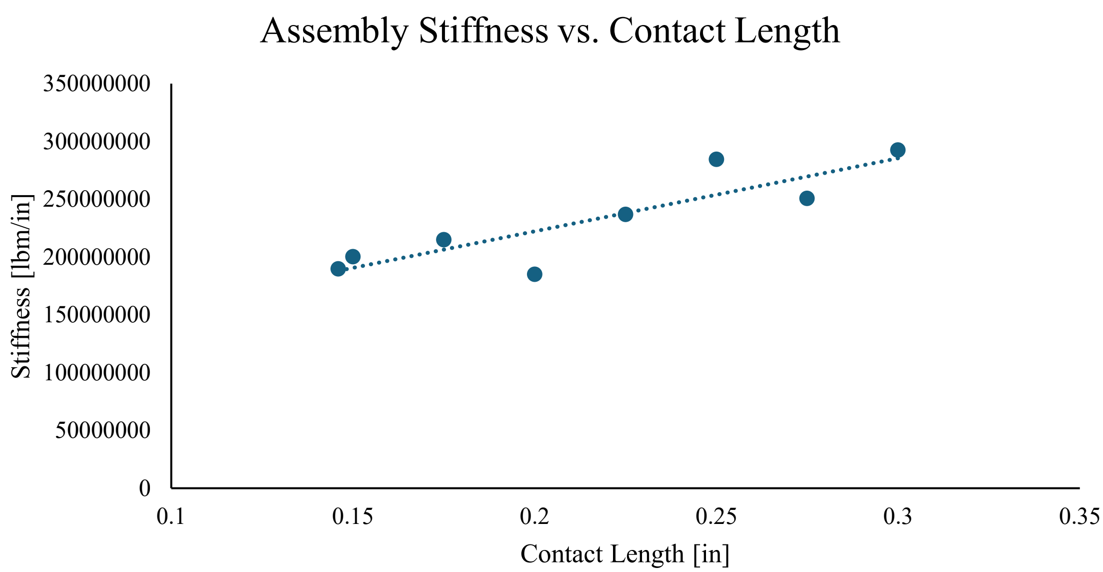


Figure 3: Increasing the pilot interference contact length gave a much stronger trend of increased assembly stiffness as more contact length became present.

This result indicates that distributed contact area contributes more effectively to global stiffness than simply increasing interference magnitude. During the pilot contact length study, dowel and pilot interferences were maintained within light interference ranges to isolate the effect of contact length. The stiffness versus interference plots clearly demonstrated that contact length is the dominant parameter influencing assembly rigidity within the ranges tested.

Although finite element analysis provides strong predictive capability, experimental verification is required to validate results. Two complementary testing methods will be used. The first method involves modal impact hammer testing. The assembly will be mounted to a rigid base consistent with the boundary conditions applied in the finite element model. Accelerometers will be attached at strategic locations on the plates. The assembly will be struck using a calibrated modal impact hammer that records input force. The resulting acceleration response will be processed using a spectral analyzer to extract frequency response functions. Peaks in the frequency spectrum correspond to natural frequencies. Measured frequencies will be compared directly with finite element predictions, and relative accelerometer motion will be compared with predicted mode shapes.

The second method involves shake table testing. The assembly will be mounted to the shake table and subjected to a frequency sweep across a specified range. Accelerometers will measure the dynamic response of the system as excitation frequency increases. Resonance peaks will be identified from the response data. Extracted natural frequencies will again be used to calculate stiffness by multiplying the squared frequency by assembly mass. Agreement between experimental and analytical stiffness values will validate the finite element model and confirm that interference induced preload is being accurately represented.

Two configurations were selected for physical testing to maximize observable stiffness differences. The first configuration represents a lower preload condition with relatively small diametral interferences and shorter pilot contact length. The second configuration represents a higher preload condition with increased dowel interference, increased pilot interference, and extended pilot contact length. Selecting both low and high preload states ensures that measurable differences in stiffness should be detectable during experimentation. These configurations correspond directly to specific design points evaluated in the finite element study, allowing direct analytical and experimental comparison.

In summary, the final design integrates vibration theory, parameterized finite element modeling, standards-based interference selection, and laboratory validation to quantify stiffness variation in rotor plate assemblies. The scaled assembly preserves critical contact mechanics while satisfying laboratory constraints. Preloaded modal analysis provides stiffness predictions grounded in fundamental physics. Experimental testing using both modal impact and shake table methods ensures verification of analytical results. This comprehensive approach provides Solar Turbines with validated insight into how interference fits and pilot geometry influence rotor plate stiffness and supports informed design decisions for future turbine assemblies.

Design Justification

Given the high tolerance nature of producing turbine engine rotor blades, we plan on minimizing the risk of tolerance stack up and propagation by milling our parts on a CNC mill. The methods for which we will test whether our physical model is within our decided tolerance is by using a CMM machine that can measure dimensions with a tolerance of ±0.0001. We will ensure that our physical model matches our CAD models using key dimensions. To do this, we will verify that the material properties of each model are as similar as possible by sourcing our stock from the same supplier and batch. We will also ensure proper machining by using the same machine, tools, and procedure. Utilizing these tools and processes, we are confident that we will be able to produce identical protype models machined to our desired dimensions. We will then test each model on the shake table and verify that the mode shapes and natural frequencies align with anticipated results.

Table 1. Justification that our design will meet each of our specifications.

#	Spec	Specification Description	Justification
1	Rotor CAD model delivered	Deliver complete CAD for female and male plates that characterize the mating surface between compressor rotor plates and aft weldment. Fully constrained dimensions	Establishes the controlled geometry baseline for analysis, fabrication and testing.
2	Material Property definition	Provide accurate material properties used for analysis with datasheet and verification	Guarantees analysis inputs are traceable and repeatable.
3	Compressor Assembly CAD Package	Deliver full compressor assembly CAD that accurately characterizes the assembly. Supports bill of materials.	Allows for interface checks. Ensures proper part coupling and part count.
4	Ansys FEA Model	Build an Ansys model that can mesh and solve our chosen geometry based on assigned loads and boundary conditions	Provides an analytical tool to predict performance and guide design configurations and alteration
5	Prototype fabrication	Manufacture a prototype to drawing specifications with inspection results	Provides a physical article for dimensional testing and validating manufacturing plan
6	Modal Simulation	Create a model that can accurately calculate the natural frequency of a choice part	Predict dynamic behavior and supports design decisions before testing
7	Modal impact test	Conduct modal impact testing and report natural frequency and mode shapes.	Validates real world dynamics and suppliers' data to our experimental calculations. Also tests and verifies our testing procedure.
8	First Natural Frequency Requirement	Demonstrate first natural frequency meets a minimum of 186 Hz.	Prevents resonance issues and confirms stiffness meets system requirements
9	Test to analysis correlation	Quantify correlation between physical model and theoretical model values. Ensure values are within predefined tolerances	Confirms the model is predictive and supports confident design iterations.
10	Manufacturing process definition	Provide a tested manufacturing plan for each part and assembly	Reduces manufacturing error, streamlines production and reduces cost.

Manufacturing Plan

Table 2 shows a summary of the manufacturing plan for our three assemblies. A full detailed build plan for the assemblies can be found in Appendix E. The two chosen configurations for the interference fit assemblies will be manufactured using the same process. The female and male plate will both be machined via a 3-axis CNC mill in the Cal Poly Aero Hangar. The first operation will require four heavy duty vise jaw grips to hold the workpiece; the second operation will require soft jaws to be machined to hold the workpiece. One to two pairs of soft jaws will be machined to be reused for assembly. Steel tabs for mounting to a shake table will be cut using the waterjet in Mustang 60. Due to the water jet's poor tolerance, the holes on the tabs will be undersized and drilled on the drill press in the Aero Hangar. Four tabs will be waterjet and welded to the female plate as it is the bottom plate that will be directly on top of a shake table. A welding jig will be 3D printed to ensure that the tabs are welded onto the female plate in the correct positions to match the hole pattern on the shake table.

Before assembling, each male and female plate will be inspected on the CMM in the IME department to assess the critical interference dimensions. For the assembly of the interference fit configurations, a temperature gradient is necessary to assemble interference fits. We plan on utilizing an oven in the MATE department with the help of Professor Beaton. First, the male plate will be placed in the oven so that the dowel pin holes will expand. Meanwhile, the dowel pins will be placed in liquid nitrogen for shrinkage. The dowel pins will then be pressed into the hot male plate. Then, the female plate will be placed in the oven to expand. The male plate will then be pressed into the female plate using an arbor press.

The welded assembly has a similar but shorter manufacturing process. Both plates will be machined on the 3-axis CNC mill in the Aero Hangar. But the process excludes machining the dowel pin holes and pilot ledges. The plates will be MIG welded together.

Table 2. Manufacturing summary.

Subsystem	Component (Part Number)	Purchase (P) Modify (M) Build (B)	Raw Materials Needed	Where/ how procured?	Equipment and Operations
2 Interference Fit Configurations	Male plate (1100, 2100)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Female plate (1200, 2200)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Dowel pins (1300, 2300)	P	3/16" OD alloy steel dowel pin	Sponsor to purchase from McMaster-Carr	n/a
	Tabs (1110, 2110)	B	0.25" thick 1026 steel	Sponsor to purchase from McMaster-Carr	waterjet to cut profile, drill press to create hole, weld to attach to plate
	Shake table mounting bolts (1111, 2111)	P	3/8" OD, 1" length Grade 8 steel bolt	Sponsor to purchase from McMaster-Carr	n/a
1 Welded Configuration	Male plate (3100)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Female plate (3200)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Tabs (3110)	B	0.25" thick 1026 steel	Sponsor to purchase from McMaster-Carr	waterjet to cut profile, drill press to create hole, weld to attach to plate
	Shake table mounting bolts (3111)	P	3/8" OD, 1" length Grade 8 steel bolt	Sponsor to purchase from McMaster-Carr	n/a
Manufacturing tools	Soft jaws	B	6061-Al machinable 8”x2.5”x1” billet	Sponsor to purchase from McMaster-Carr	CNC mill – mill the profile of the machined plate
	Welding jig	B	PLA filament	Makerspace	Bambu printer

Design Verification

The design verification, or physical assemblies, of this project will be used to verify the Ansys model created to study the effects radial interference fits from the dowel pins and pilot fits have on overall system stiffness. As the Ansys model will be made to study many different conditions, three physical test assemblies will be manufactured to ensure the Ansys model is providing accurate results. Many specific checks will be run on the manufactured parts to ensure each part was made to the correct and necessary specifications for testing. Table 3 below includes these specification tests. Before the physical modal analysis is completed on the assemblies, multiple dimensional analyses are performed to import dimensions into the Ansys model that correctly match physical build. These assemblies will be used to verify the validity of the Ansys model; therefore, multiple tests of product sizing will be conducted before any other testing.

Table 3. Design Verification Plan

Project:		6RL-B	Sponsor:	Manny Aldana			Date:	2/13/26
TEST PLAN								
Test #	Spec #	Test Description	Measurements	Acceptance Criteria	Facilities/ Equipment	Parts Needed	Member	TIMING Start date
1	1	Open, rebuild, and validate male and female plate models. Confirm mating surfaces and constraint status	Use rebuild/error log to check current status to ensure models are fully defined with key interface dimensions	Zero rebuild errors, all sketches are fully constrained.	SolidWorks	Digital CAD files	TBD	TBD
2	2	Confirm properties match datasheet and are properly imported into Ansys	Elastic modulus, Poisson's ratio, density, yield strength	Required properties are provided with units and source	Materials datasheet	none	TBD	TBD
3	3	Open assembly, check mates and interference values. Compare assembly BOM with project BOM	Use rebuild/error log, check DOF, and interference values	Opens with zero errors, interference values match, BOM match	SolidWorks	Digital CAD files	TBD	TBD

4	4	Import geometry and ensure proper meshing, correct loading values, and correct load locations are generated. Run a simulation properly	Mesh count, mesh type, solver completion status, run logs, boundary conditions, documented loads	Model meshes successfully, model completes successful simulation, values generated are within reason	Ansys	Digital CAD files	TBD	TBD
5	5	Manufacture prototype per drawing dimensions and manufacturing procedure. Inspect critical dimensions and record material certifications	Dimensional inspection using calipers and CMM. Check critical dimensions	All critical dimensions withing drawing tolerance. Material properties match specifications. No major damage	CMM machine, Calipers, data sheets	Machined parts, drawing, fasteners,	TBD	TBD
6	6	Run modal analysis on selected part using defined boundary conditions and calculate mode shapes and verify they are within reason	Mode shapes, solvers settings, Boundary conditions definition, Mass and stiffness values	Modal analysis runs completely. Output mode shapes and natural frequencies.	Ansys model	Material data sheet	TBD	TBD
7	7	Perform impact hammer test, calculate/measure frequency response.	Utilize frequency response to determine measurement alignment.	Test produced data for coherence. Ensure data has/reduces noise	Shake table, systems laboratory, modal impact hammer	Prototype assembly, impact hammer, accelerometers	TBD	TBD
8	8	Confirm first natural frequency meets requirements under defined configurations.	experimental first natural frequency match Configuration and boundary conditions	Pass/fail: first natural frequency must be greater than 186 Hz	Shake table, systems laboratory, modal impact hammer	Prototype assembly, impact hammer, accelerometers	TBD	TBD

9	9	Compare model test results with FEA results for agreement. Quantity error	Percent difference per model between test and theoretical values.	All defined values are within +- 15 percent. and are recorded	Ansys	FEA results, test results	TBD	TBD
10	10	Test the processing and manufacturing plan.	Verify tooling list, operations order, and operation time	At least one prototype build is executed using manufacturing and processing plans.	CNC mill, aero hanger	Drawings, Prototype manufacturing plans, inspection plan	TBD	TBD

Budget

The budget spending for this project is driven solely by the testing and verification assembly that will be used to verify the Ansys model. This test and verification assembly, also called the verification prototype, is made up of three sub-assemblies that represent three different configurations to be tested. Table 4 shows a list of materials that will be purchased to manufacture these three sub-assemblies. Most of the funds will be used for raw steel stock, with the remaining funds being used for tooling and purchased mechanical parts. This table does not include the exhaustive list of tooling needed to manufacture the assemblies. Other tooling and parts that were listed in the manufacturing plan will be provided by Cal Poly's machine shops. Although the project is under budget so far, it is predicted that there will be a need to order more stock or tooling as the process progresses through the remainder of winter quarter.

Table 4. Budget summary.

Item Description	Vendor	Vendor's Part #	iBOM #	Total Cost	Purchase Status
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	1100	$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	1200	$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	2100	$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	2200	$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	3100	$74.87	Standby
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	3200	$74.87	Standby
Dowel Pins (3/16''D)	McMaster-Carr	98372A117	1300	$31.26	On the way
Carbide Drill Bit (11/64)	McMaster-Carr	27515A42	-	$23.03	On the way
Reamer (0.1865)	McMaster-Carr	3087A52	-	$25.95	On the way
Reamer (0.1855)	McMaster-Carr	2995A212	-	$30.46	On the way
Soft Jaws	McMaster-Carr	1005N35	-	$64.27	On the way
Plate Tab Stock (12''x12''x1/4'')	McMaster-Carr	6544K25	-	$84.38	On the way
Bolts (3/8''-16, 1''L)	McMaster-Carr	92620A624	-	$13.68	On the way

Total Budget:	$4000
Amount Spent:	$618.31
Planned Spending:	$1000
Remainder:	$3000

Appendix A: Design Analysis

Many calculations were performed to analyze the feasibility of the physical test models. With the given goals of the project, the team calculated the predicted hoop stress of the radial and axial interference fits, the resistive force capabilities of the radial fits combating the axial interference fits, the required bolt size to maintain the axial interference requirements, and more. An example of this can be seen in Figures A-1 and A-2. Overall, the decision to remove the study of axial interference on the plates was made after said calculations predicted the torque specs to be beyond the manufacturing capabilities of the team given the tools available. This can be seen in Figure A-3 where we calculated the tensile force the bolt would need to exert to create the required axial interference and its subsequent preload and torque specification. Further analysis of this design was completed in Ansys and is explained in further detail in the Ansys portions of this report.

Config. 1		
Input	Value	Units
A	3.5	in
B	2.345	in
C	2.342	in
D	1	in
E	1	in
F	0.83	in
G	0.833	in
H	0.1855	in
I	0.1875	in
J	0.5	in
E_o	2.97E+07	psi
E_i	2.97E+07	psi
v_o	0.29	-
v_i	0.29	-
μ	0.74	-
# of Dowel Pins:		5
k	0.50	scaled version
Length of shear	0.837	

Manny recommended Interferences			
Interf.	Value	Units	
δ_{PR}	0.003	in	
δ_{PA}	-0.001	in	solars tolerance
δ_{DR}	0.002	in	radial

Resulting Surface Areas			
SA	Value	Units	
PR	2.457441738	in^2	
PA	21.20881346	in^2	outer area
DR	0.294524311	in^2	
PA_2	17.23152261	in^2	face area
PA_3	6.572211831	in^2	face cut area

	SciNot	Normal	Units
Pilot Radial Pressure:	1.05E+04	10504.97	psi
Dowel Radial Pressure:	3.14E+05	314015.63	psi

net Forces		
Force	Value	Units
Pilot Radial	25815.4	lbs
Dowel Radial	462426.2	lbs

Resulting Frictional Force: in axial direction		
Radial Fits:	361298.7334	lbs

Config 2.		
Input	Value	Units
A	3.5	in
B	3.345	in
C	3.342	in
D	1	in
E	1	in
F	0.83	in
G	0.831	in
H	0.1855	in
I	0.1875	in
J	0.5	in
E_o	2.97E+07	psi
E_i	2.97E+07	psi
v_o	0.29	-
v_i	0.29	-
μ	0.74	-
# of Dowel Pins:		5
k	0.38	scaled version
Length of shear	0.837	

Manny recommended Interferences			
Interf.	Value	Units	
δ_{PR}	0.003	in	
δ_{PA}	-0.001	in	solars tolerance
δ_{DR}	0.003	in	radial

Resulting Surface Areas			
SA	Value	Units	
PR	3.548730495	in^2	
PA	3.333151266	in^2	outer area
DR	0.294524311	in^2	
PA_2	35.08833525	in^2	face area
PA_3	7.432615519	in^2	face cut area

	SciNot	Normal	Units
Pilot Radial Pressure:	1.18E+03	1176.37	psi
Dowel Radial Pressure:	4.71E+05	471023.44	psi

net Forces		
Force	Value	Units
Pilot Radial	4174.6	lbs
Dowel Radial	693639.3	lbs

Frictional Force: in axial direction		
Radial Fits:	516382.2868	lbs

axial interference		
pressure	Value	Units
Pilot axial	-3.57E+04	psi
Dowel Radial	0.0	lbs

axial force from interference		
Force	value	units
pilot axial with PA_2	-6.144E+05	lbf
pilot axial with step/PA_3	-2.343E+05	lbf

compare froces	
face	dowel
-2.34E+05	3.61E+05

good

bolt resultant

1.27E+05 lbf

126971.37

axial interference		
pressure	Value	Units
Pilot axial	-3.57E+04	psi
Dowel Radial	0.0	lbs

axial force from interference		
Force	value	units
pilot axial with PA_2	-1.254E+06	lbf
pilot axial with step/PA_3	-2.656E+05	lbf

compare froces		
face	dowel	
-2.66E+05	5.16E+05	lbf

good

bolt resultant

2.51E+05 lbf

250740.07

Figure A-1: Example of axial vs radial interference fit forces comparison. Both configurations determine that the frictional force of the dowel pin from radial interference is enough to combat the axial interference of the pilot fit based on the dimensions given.

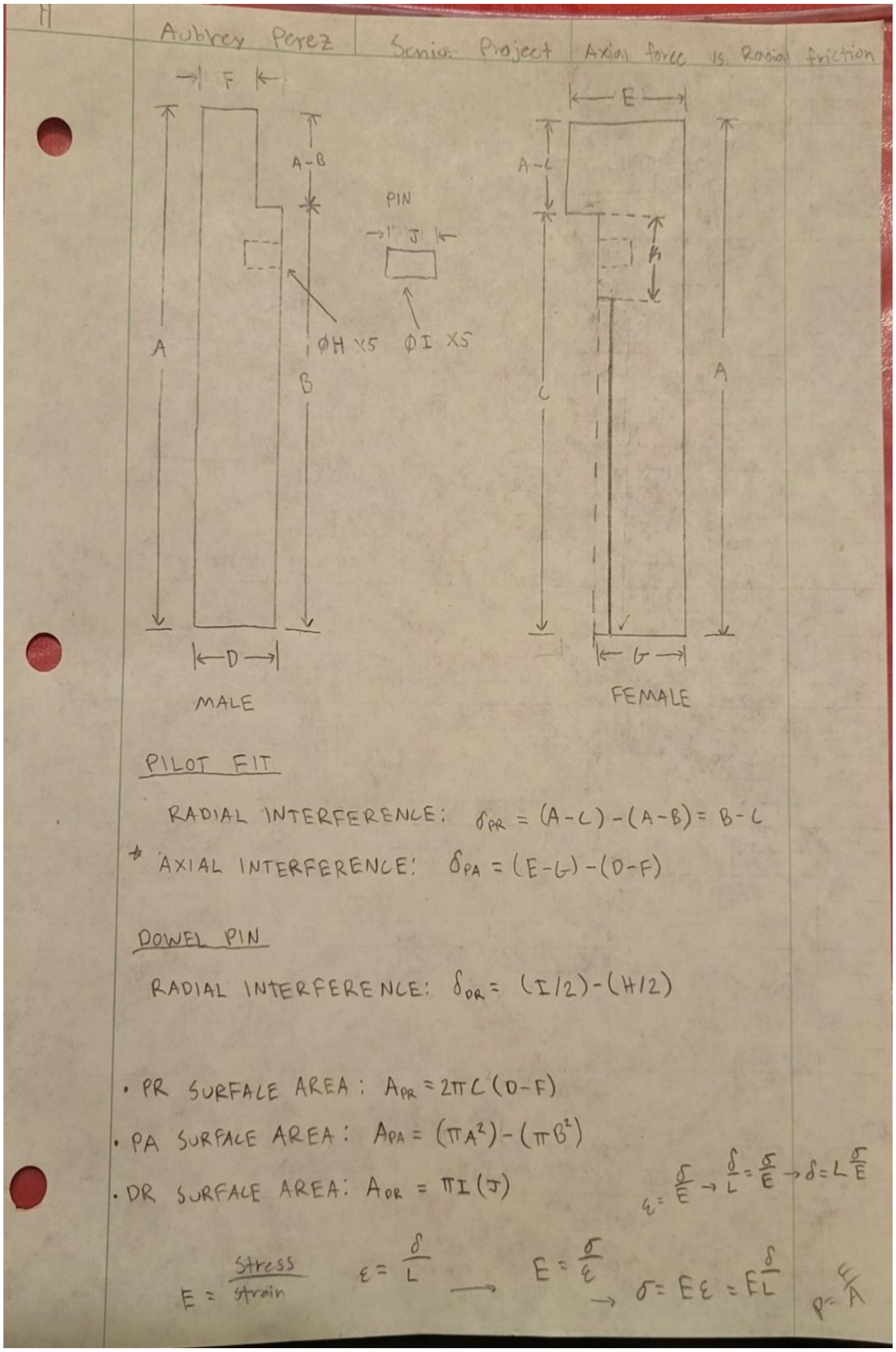
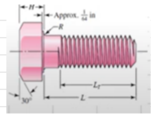


Figure A-2: Example dimensioning set up used for interference calculations, such as the ones shown in Figure A-1.

clamp meber stress calculator		Key	inputs	calculated solutions
4 plates 2 washers and holes				
hardware:				
Bolt	1.25, unc = 7, unf = SAE grade 8			
Max tensile force =	1.45E+05	145350		
nut height	1.16	inches		
thread	7	threads per inch		
washer diameter theory	1.875	inches		
choicen washer diamter	2.5	inches		
bolt lengt calc	6.168571429	inches		
force trying to keep	9.15E+04	lbf	91455.3	



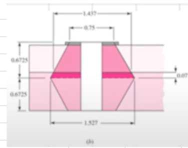


Table 8–17
Fully Corrected Endurance Strengths for Bolts and Screws with Rolled Threads*

Grade or Class	Size Range	Endurance Strength
SAE 5	$\frac{1}{4}$–1 in	18.6 kpsi
	$1\frac{1}{8}$–$1\frac{1}{2}$ in	16.3 kpsi
SAE 7	$\frac{1}{4}$–$1\frac{1}{2}$ in	20.6 kpsi
SAE 8	$\frac{1}{4}$–$1\frac{1}{2}$ in	23.2 kpsi
ISO 8.8	M16–M36	129 MPa
ISO 9.8	M1.6–M16	140 MPa
ISO 10.9	M5–M36	162 MPa
ISO 12.9	M1.6–M36	190 MPa

*Repeatedly applied, axial loading, fully corrected.

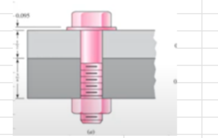


Table 8–16
Fatigue Stress-Concentration Factors K_f for Threaded Elements

SAE Grade	Metric Grade	Rolled Threads	Cut Threads	Fillet
0 to 2	3.6 to 5.8	2.2	2.8	2.1
4 to 8	6.6 to 10.9	3.0	3.8	2.3

Material props			
	E steel	3.00E+07	psi
	E cast iron,imu,	1.45E+07	psi
	E alumin	1.03E+07	psi

units inches

member	thickness		elastic modulus
top washer	0.165	in	3.00E+07
top cup	0.75	in	3.00E+07
top void	0.25	in	0.00E+00
male plate	1	in	3.00E+07
void 2	0.25	in	0.00E+00
female plate	1	in	3.00E+07
void3	0.25	in	0.00E+00
bottom cup	0.75	in	3.00E+07
bottom washer	0.165	in	3.00E+07

FATIGUE 4 pressure vessel		
Max pressure	5.50E+05	psi
Min pressure	0	psi
pressure dia	0.8	in^2
#BOLTS	1	
S_e	1.86E+04	psi

need to update value of D for meteric problems, or undo

For solid metal, with hole

	frusta dimensions					
	segment	thickness	Diamteter	E	K	1/k
washer	1	0.165	1.875	3.00E+07	3.287E+08	3.04226E-09
t cup	2	0.75	2.065526	3.00E+07	1.384E+08	7.22771E-09
void	3	0.25	2.931026	0.00E+00		1.32308E-07
t male	4	1	3.219526	3.00E+07	2.937E+08	3.40523E-09
void	5	0.125	4.373526	0.00E+00		2.99792E-07
void	6	0.125	4.517776	0		0
female	7	1	3.219526	3.00E+07	2.937E+08	3.40523E-09
void	8	0.25	2.931026	0		1.32308E-07
cup2	9	0.75	2.065526	3.00E+07	1.384E+08	7.22771E-09
washer	10	0.165	1.875	3.00E+07	3.287E+08	3.04226E-09
	thickness checl	4.58		member stiffnes		1.690E+06

t_tw = thickness top washer

net grip length	4.58	
half grip	2.29	
calculated bolt length	6.168571429	
rounded for convience	4.75	

pressure area	0.502654825	in^2
Net max load calc	2.76E+05	lbf
net max load used	9.15E+04	
net min load	0.00E+00	
max force per bolt plate assembly calc	9.15E+04	newtons
max force per bolt plate assembly chosen	9.15E+04	
min force per bolt(P)	0.00E+00	

$(2t\tan\alpha + D + d)(D - d)$

With $\alpha = 30°$, this becomes

$$k = \frac{0.5774\pi E d}{\ln\dfrac{(1.155t + D - d)(D + d)}{(1.155t + D + d)(D - d)}}$$

bolt properties			
bolt elastic modulous	3.00E+07	psi	
bolt length choicen	4.75	in	
bolt diameter	1.25	in	P_max
Strength _Ultimate	1.50E+05	psi	P_min
S_y (yeild)	1.30E+05	psi	
s_p (Proof) calculated	1.11E+05	psi	
s_p (Proof) stated	1.10E+05	psi	
friction factor	0.2		

load amplitude	4.57E+04	newtons
mean amplitude	4.57E+04	
Stress		
pre-stress	5.50E+04	
stress amplitude	3.80E+10	
mean stress		

Dimensions	in	in
threaded length (8.13)	2.75	in
threaded length (8.14)	8.5	mm
unthreaded length	2	in
threaded length in grip	2.580	in
norminal area	1.22718463	in^2
threaded area	0.969	in^2

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$$

$$L_T = \begin{cases} 2d + \frac{1}{4}\text{ in} & L \le 6\text{ in} \\ 2d + \frac{1}{2}\text{ in} & L > 6\text{ in} \end{cases} \quad (8\text{–}13)$$

$$L_T = \begin{cases} 2d + 6 & L \le 125 \quad d \le 48 \\ 2d + 12 & 125 < L \le 200 \\ 2d + 25 & L > 200 \end{cases} \quad (8\text{–}14)$$

bolt stiffness	6.989E+06	lbf/in	
C (joint complaince)	0.805		
preload F_i	5.330E+04	53295.000	lbf
F_b (F bolt exp	1.269E+05	126943.470	
torque	1.332E+04	13323.75	in-lbf
eta_p = proof bolt FOS	8.397E-01		
eta_not (opening FOS)	2.993E+00		
eta_L (proof load)	7.236E-01		
what bolt can take	1.45E+05	145350	
copy and paste equation not cells			

we need $F_i > 60\% F_p$
to prevent backing out

Did a sanity check, looks good to me

spare(if acciedent)/check	
threaded length (8.13) EQ	2.75
threaded length (8.14)	8.5

Figure A-3: Bolt calculations for the case of needing a center bolt through the assembly to maintain the axial interference of the pilot fit.

Appendix B: Gantt Chart

The Gantt chart followed for this Critical Design Review can be seen below in Figure B-1. The majority of the documentation and milestones leading up to this design review can be found in this report. Since the Preliminary Design Review, a final Ansys model and testing design was created to achieve the end project goal. Through the months of January and February, the team created a manufacturing plan, testing and verification plan, budget and spending table, and detailed CAD to explain the final decisions made by the team for this project. The next phase in the project after this phase is Safety and Verification planning, which includes studying the safety of the planned testing methods and making alterations if required.

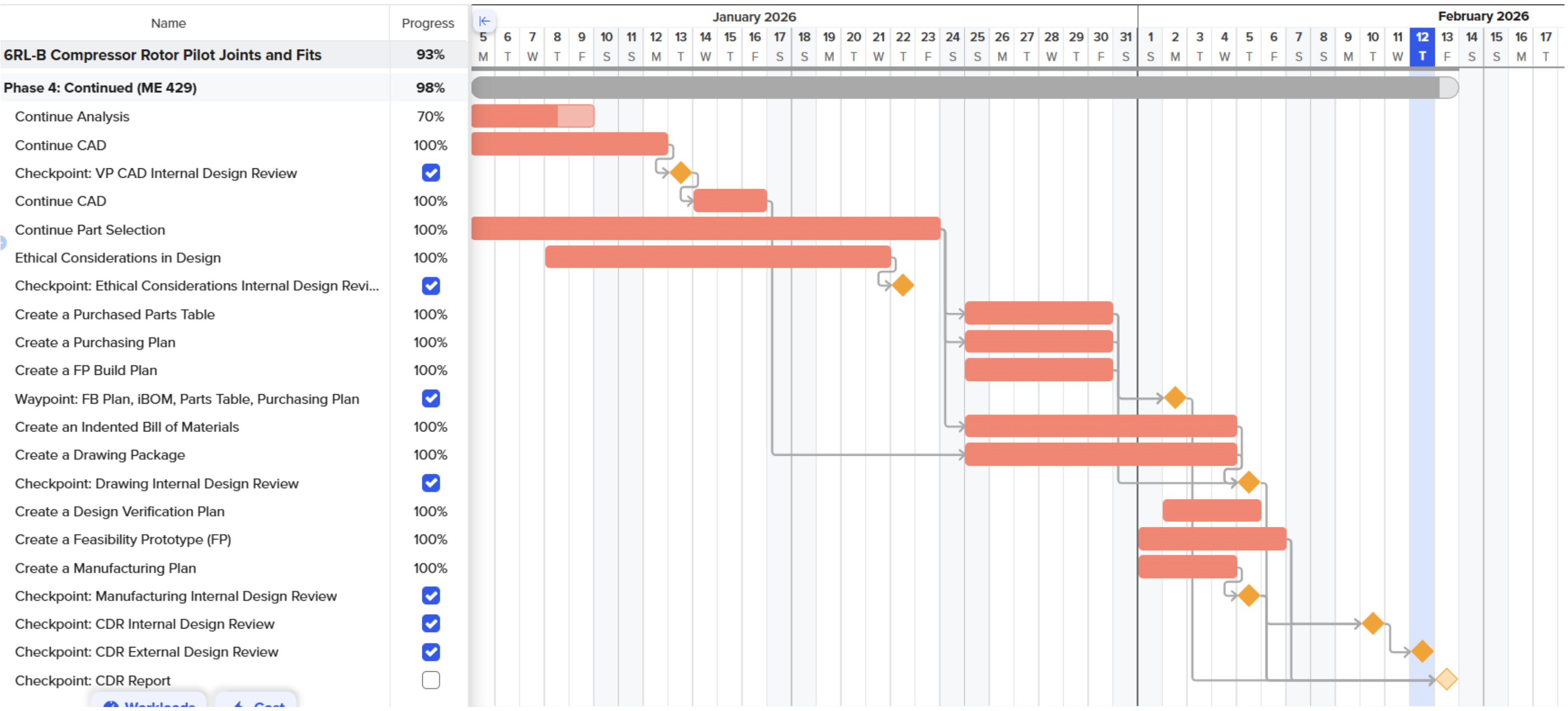


Figure B-1: Phase 4 Gantt chart leading up to Critical Design Review.

Appendix C: FMEA

Along with designing the Ansys model and physical testing assembles, a failure mode and effects analysis (FMEA) was conducted to study ways the physical assemblies could fail. Most of the failure modes identified are preventable with careful manufacturing and frequent measurement analysis to make sure no complications arise. Nonetheless, the main points of potential failure are always a concern. These failure modes are listed below in Table C-1 along with actions to prevent such complications.

Table C-1: FMEA table of foreseeable failure modes during manufacturing of the testing assemblies. Due to having no high-RPN failure modes, no additional actions than these were written.

System / Function	Potential Failure Mode	Potential Effects of the Failure Mode	Severity	Potential Causes of the Failure Mode	Current Preventative Activities	Occurence	Current Detection Activities	Detection	RPN
securely mount to shake table	tabs deform/detach	a) invalid data due to incorrect boundary conditions b) unsafe testing	6	1) poor weld penetration 2) waterjet tolerance error for holes	1) utilizing a welding jig 2) weld inspection 3) accurate drilling	3	visual weld inspection	2	36
maintain pilot radial interference	interference fit too large	a) cracking during press b) yielding c) residual stress deformations	8	1) machine tolerance stackup 2) holes didn't expand fully in the oven	1) CMM inspection 2) thermal expansion calculation 3) tolerance analysis	2	dimensional inspection, monitoring press fit	2	32
gather accurate modal data	incorrect accelerometer data	a) incorrect natural frequencies b) seemingly invalid ANSYS model c) incorrect conclusion of stiffness correlation	8	1) poor accelerometer mounting 2) cable interference 3) calibration error	1) proper mounting with threaded holes or glue 2) purchase very light accelerometer 3) lab procedure review	3	repeatability check, logic check	3	72
maintain concentric alignment	misalignment between male/female plates	1) eccentricity introduced in system 2) false modal response 3) out-of-plane modes	8	1) uneven heating in oven 2) dowel pin tilt 3) arbor press misalignment	1) controlled heating time 2) alignment fixture/method 3) concentricity inspection on CMM	3	CMM inspection post assembly, dial indicator runout check	3	72

Appendix D: Design Hazard Checklist

Along with the FMEA completed to find concerning failure points, a list of generic mechanical hazards was studied. From that list, the team analyzed if any of these hazards were applicable to this project. The list of hazards and applicability to this project can be seen below in Table D-1.

Table D-1: Table of generic mechanical hazards. The leftmost column explains whether the specific hazard is applicable to this project, and therefore something the team will need to take greater caution with.

Y	N	Category
		1. Will any part of the design create hazardous revolving, reciprocating, running, shearing, punching, pressing, squeezing, drawing, cutting, rolling, mixing or similar action, including pinch points and sheer points?
		2. Can any part of the design undergo high accelerations/decelerations?
		3. Will the system have any large moving masses or large forces?
		4. Will the system produce a projectile?
		5. Would it be possible for the system to fall under gravity, creating injury?
		6. Will a user be exposed to overhanging weights as part of the design?
		7. Will the system have any sharp edges?
		8. Will any part of the electrical systems not be grounded?
		9. Will there be any large batteries or electrical voltage in the system above 40 V?
		10. Will there be any stored energy in the system such as batteries, flywheels, hanging weights or pressurized fluids?
		11. Will there be any explosive or flammable liquids, gases, or dust fuel as part of the system?
		12. Will the user of the design be required to exert any abnormal effort or physical posture during the use of the design?
		13. Will there be any materials known to be hazardous to humans involved in either the design or the manufacturing of the design?
		14. Can the system generate high levels of noise?
		15. Will the device/system be exposed to extreme environmental conditions such as fog, humidity, cold, high temperatures, etc?
		16. Is it possible for the system to be used in an unsafe manner?
		17. Will there be any other potential hazards not listed above? If yes, please explain the reverse.

Description of Hazard	Planned Corrective Action	Planned Date	Actual Date
During the shake table tests, the assembly can be exposed to high lateral accelerations. Under strong lateral forces, there is a potential danger of the momentum of the assembly breaking the mounting tabs and becoming unconstrained.	To prevent this from happening, it will be ensured that the person planning to weld the mounting tabs to the plate will practice MIG welding to ensure proper weld penetration. A visual weld inspection test will be performed before each shake table test. The mounting tab holes will also be inspected to confirm they are not too large where the mounting holes would slide around during the shake table test.	Weld practice starting: 2/23/26 Weld and tab inspections during assembly practice	
Though not a strong projectile, there is a possibility that the interference fit pops the plates apart, which is a safety concern.	To prevent this, it is planned to alter the design to include a couple of bolts and nuts to loosely constrain the plates together in the case that they pop apart. The bolts will not tightly constrain the plates together due to that affecting the testing resulting.	2/20/26	
The assembly is considerably heavy and would need to be transported to and from the vibrations lab. Dropping the assembly could cause injuries.	To prevent this, a transportation method involving crates/boxes that at least two people carry will be used. These boxes/crates will be purchased via order form.	3/6/26	
If not properly placed on the shake table, the assembly can be incorrectly constrained and be a dangerous moving object during testing.	To prevent this, a testing procedure/manual will be created to ensure that the testing is performed safely.	2/20/26	

Appendix E: Manufacturing Plan

This appendix shows a detailed plan for the manufacturing and assembly process needed for each assembly configuration. Following the first prototype that is planned to be manufactured at the end of the winter quarter, this plan will be updated as needed as better techniques and methods will be found.

Male/Female Plate

CNC Mini Mill

1. Soft Jaws
    a. Using an end mill, pocket out the profile of a 7” outer diameter circle with a depth of 0.25”.
2. 1st Operation
    a. Hold the round stock in the vise using four heavy duty vise jaw grips; the top half of the stock will be machined.
    b. Using a face mill, face off the top.
    c. Using an end mill, decrease the outer diameter of the stock to 7 inches.
    d. Using a spot drill, a reamer, and a drill, drill a through clearance hole for a 1.5” bolt.
3. 2nd Operation
    a. Flip the part over and put the machined side of the stock in the soft jaws.
    b. Using a face mill, remove material from the top of the stock to produce a thickness of 1.25” inch.
    c. Using an end mill, decrease the outer diameter of the stock to 7 inches.
    d. Using an end mill, remove material to create the pilot interference ledge (male/female ledge). Using an end mill, remove material to create the second ledge.
    e. Using a spot drill, a reamer, and a drill, drill the dowel pin holes with a diameter of 3/16” and a depth of 0.4".

Water Jet

4. Water jet the mounting tabs out of 0.25" thick 1026 steel. Undersize the water jet holes.

Drill Press

5. Drill 3/8” holes into the waterjet tabs.

3D Printer

6. Print a welding jig that will position the tabs onto the female plate and ensure the correct hole pattern.

Welding

7. Scotch Brite the tabs to weld-prep them.
8. Using the welding jig, weld the tabs onto the female plate.

CMM

Critical Dimensions to Check:

- Dowel pin hole diameter
- Dowel pin depth
- Pilot fit ledge depths and lengths
- Diameters of pilot ledges

Welded Plate

CNC Mini Mill

1. Soft Jaws
    a. Using an end mill, pocket out the profile of a 7” outer diameter circle with a depth of 0.25".
2. 1st Operation
    b. Hold the round stock in the vise using four heavy duty vise jaw grips; the top half of the stock will be machined.
    c. Using a face mill, face off the top.
    d. Using an end mill, decrease the outer diameter of the stock to 7 inches.
    e. Using a spot drill, a reamer, and a drill, drill a through clearance hole for a 1.5” bolt.
3. 2nd Operation
    f. Flip the part over and put the machined side of the stock in the soft jaws.
    g. Using a face mill, remove material from the top of the stock to produce a thickness of one inch.
    h. Using an end mill, pocket out the center.

Welding

4. Tack the two plates together.
5. Place on rotary table and rotate slowly as team member welds stationary with MIG torch.

<u>Assembly – Interference Fits</u>

1. Place the male plate into the MATE department’s oven to heat up to 350 degrees Celsius for ~45 minutes.
2. Place the dowel pins in liquid nitrogen for ~10 minutes.
3. Using an arbor press, press the dowel pins into the hot male plate.
9. Place the female plate into the MATE department’s oven to heat up to 350 degrees Celsius degrees for ~45 minutes.
10. Using an arbor press, press the male plate with the dowel pins into the hot female plate.

Appendix F: Drawing and Specifications Package

The drawing and specifications package is a group of drawings that show the planned-out manufacturing and assembly plan for the verification prototype. First, there is a copy of the iBOM for the three assemblies. The total cost for the three assemblies comes to around $400. About $200 more is estimated to be used for tooling. Around $3000 remains in the total project budget, leaving room for extra purchases if needed. Next is the Purchased Parts table that includes the list of unmodified parts used in the three assemblies. Finally, the figures are drawings of the three assemblies, including the individual parts of the assemblies, as well as the assembled views and exploded views of each assembly. Figures E-1 through E-5 are the drawings for the first two assemblies, where the plates will be connected with dowel pins and a pilot fit, while Figures E-6 through E-8 are the drawings for the third assembly, where the plates are welded together.

Cost

Compressor Rotor Press Fit Validation Assembly

Indented Bill of Material (iBOM)

Assy Level	Part Number	Lvl0	Lvl1	Lvl2	Lvl3	Lvl4	Qty	Part Cost	Total Cost	Source	URL	More Info
0		Final Assy										-
1	1000		Assembly 1									-
2	1100			Male Plate			1	$ 55.80	$ 55.80	Custom		-
3	1110				Mounting Tabs		4	$ 5.00	$ 20.00	Custom		-
4	1111					3/8"-16 Bolt	4	$ 0.55	$ 2.20	McMaster	https://wv	-
2	1200			Female Plate			1	$ 55.80	$ 55.80	Custom		-
2	1300			3/16" Dowel Pins			5	$ 0.21	$ 1.05	McMaster	https://w	-
1	2000		Assembly 2									-
2	2100			Male Plate			1	$ 55.80	$ 55.80	Custom		-
3	2110				Mounting Tabs		4	$ 5.00	$ 20.00	Custom		-
4	2111					3/8"-16 Bolt	4	$ 0.55	$ 2.20	McMaster	https://wv	-
2	2200			Female Plate			1	$ 55.80	$ 55.80	Custom		-
2	2300			3/16" Dowel Pins			5	$ 0.21	$ 1.05	McMaster	ww.mcm	-
1	3000		Assembly 3									-
2	3100			Male Plate			1	$ 55.80	$ 55.80	Custom		-
3	3110				Mounting Tabs		4	$ 5.00	$ 20.00	Custom		-
4	3111					3/8"-16 Bolt	4	$ 0.55	$ 2.20	McMaster	https://wv	-
2	3200			Female Plate			1	$ 55.80	$ 55.80	Custom		-
	Total Parts						**40**		**$ 403.50**			-

Purchased Parts Table

Part Number	Part Name	Vendor	Vendor Part Number	Price	Notes	Link
1300, 2300	3/16" Dowel Pin	McMaster	98381A505	10.37	Pack of 50 3/16" pre-cut dowel pins	https://ww -
1111, 2111, 3111	3/8"-16 Bolt	McMaster	92620A624	13.68	3/8"-16 course bolt	https://ww -

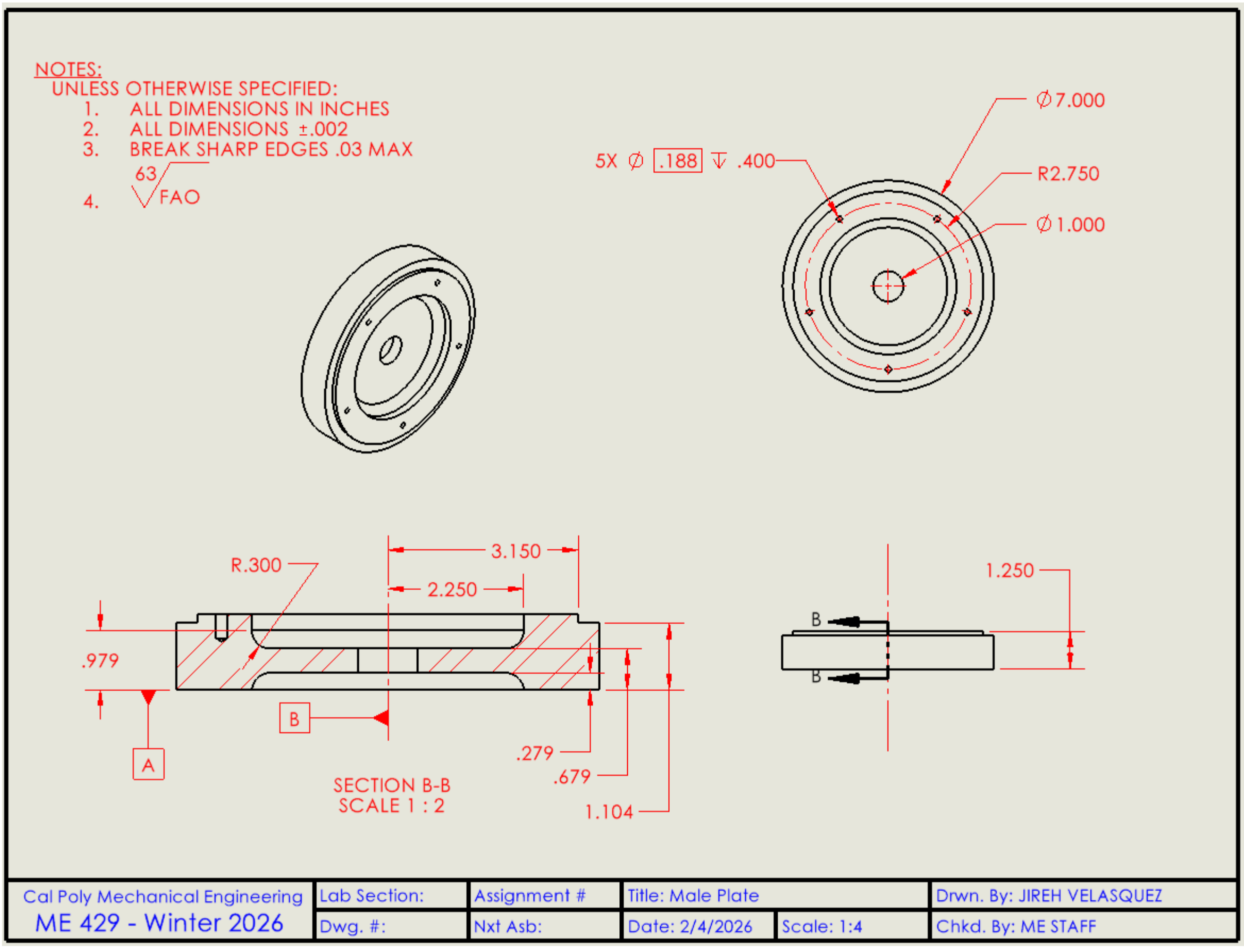


Figure E-1: Detailed drawing of the Male Plate for assemblies 1000 and 2000

Male Plate – Spec Sheet

Item	Specification
Title	Male Plate
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Jireh Velasquez
Checked by	Aubrey Perez
Date	2/4/2026
Scale	1:4 (sheet shows)
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Outer lip radius callout	R2.750
Step location (from centerline)	3.150 (to outer step), 2.250 (to inner step)
Lip/transition fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000
Hole pattern	5X Ø0.188, depth 0.400

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104

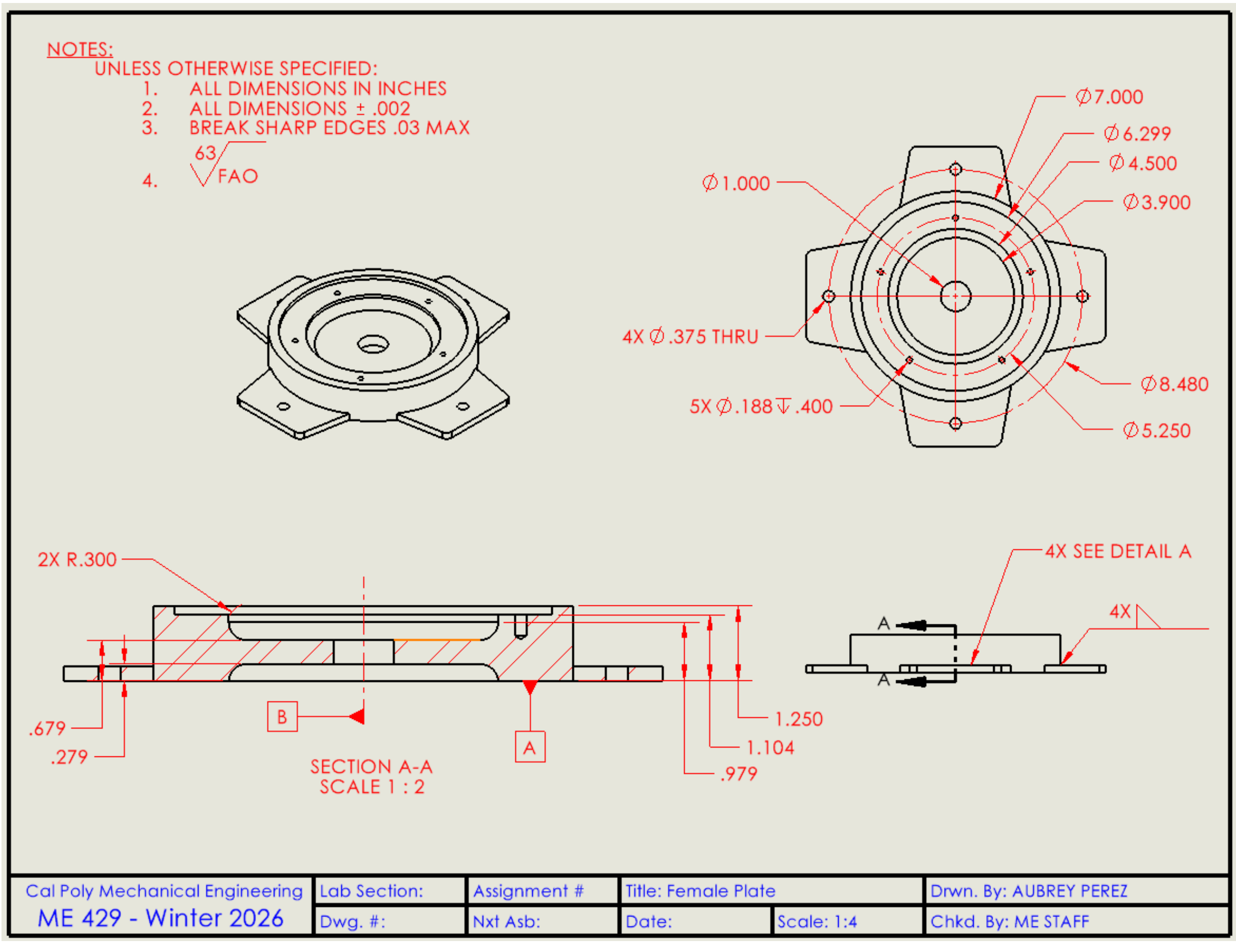


Figure E-2: Detailed drawing of Female Plate for assemblies 1000 and 2000

Female Plate – Spec Sheet

Item	Specification
Title	Female Plate
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:4
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Concentric lip/step diameters	Ø6.299, Ø4.500, Ø3.900
Lip/transition fillets	2X R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000
Thru holes	4X Ø0.375 THRU
Hole pattern (blind)	5X Ø0.188, depth 0.400
Tab outside diameter/extent	Ø8.480
Bolt circle / pattern diameter	Ø5.250

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104, 1.250

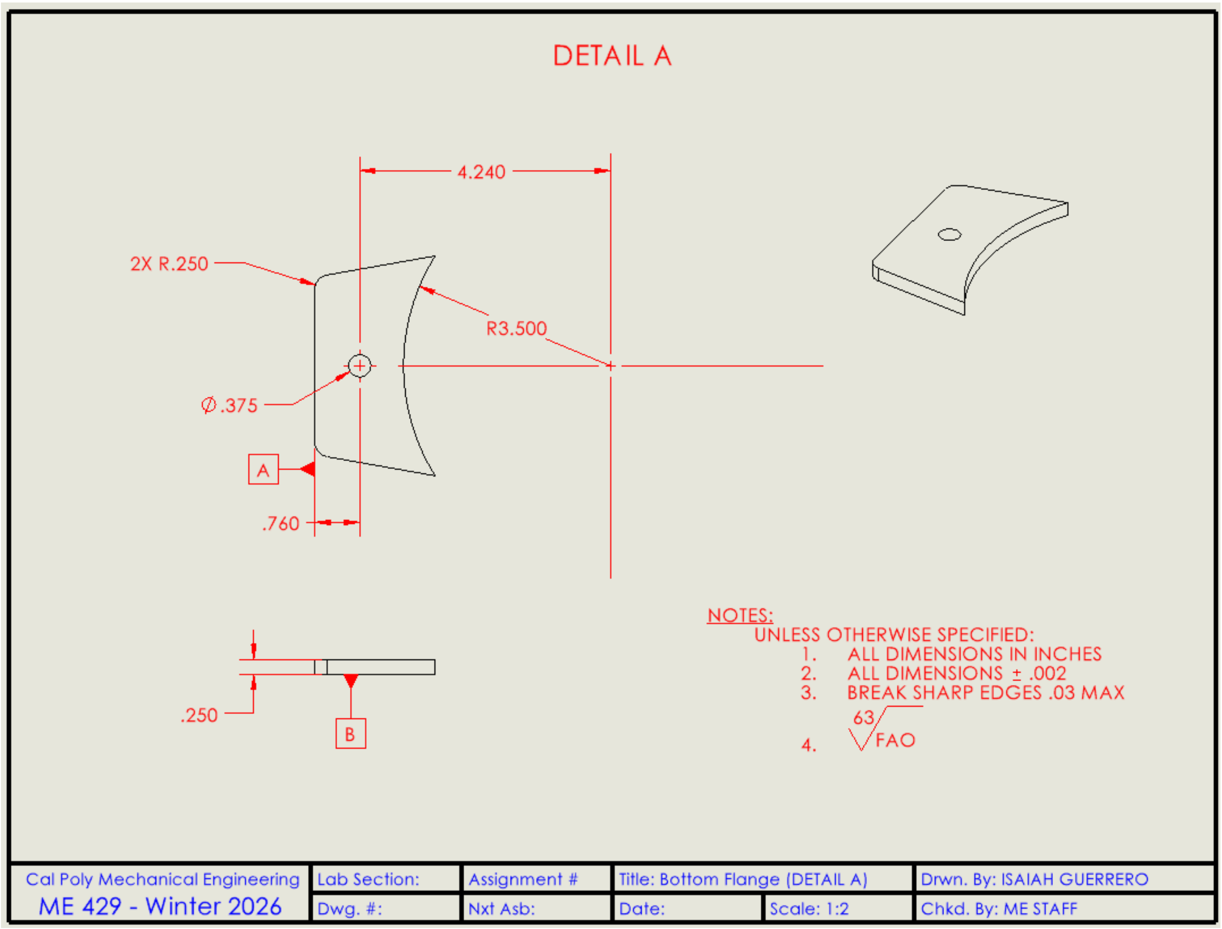


Figure E-3: Detailed drawing of the Bottom Flange attached to the Female Plate for assemblies 1000 and 2000

Bottom Flange (DETAIL A) – Spec Sheet

Item	Specification
Title	Bottom Flange (DETAIL A)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Isaiah Guerrero
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Interface note	Does not define cup lip, but affects assembled footprint and bolt-up

Other key features

Item	Specification
Overall length	4.240
Inner curve radius	R3.500
Corner/edge radii	2X R0.250
Hole	Ø0.375
Hole location offset	0.760 (as shown)
Thickness	0.250

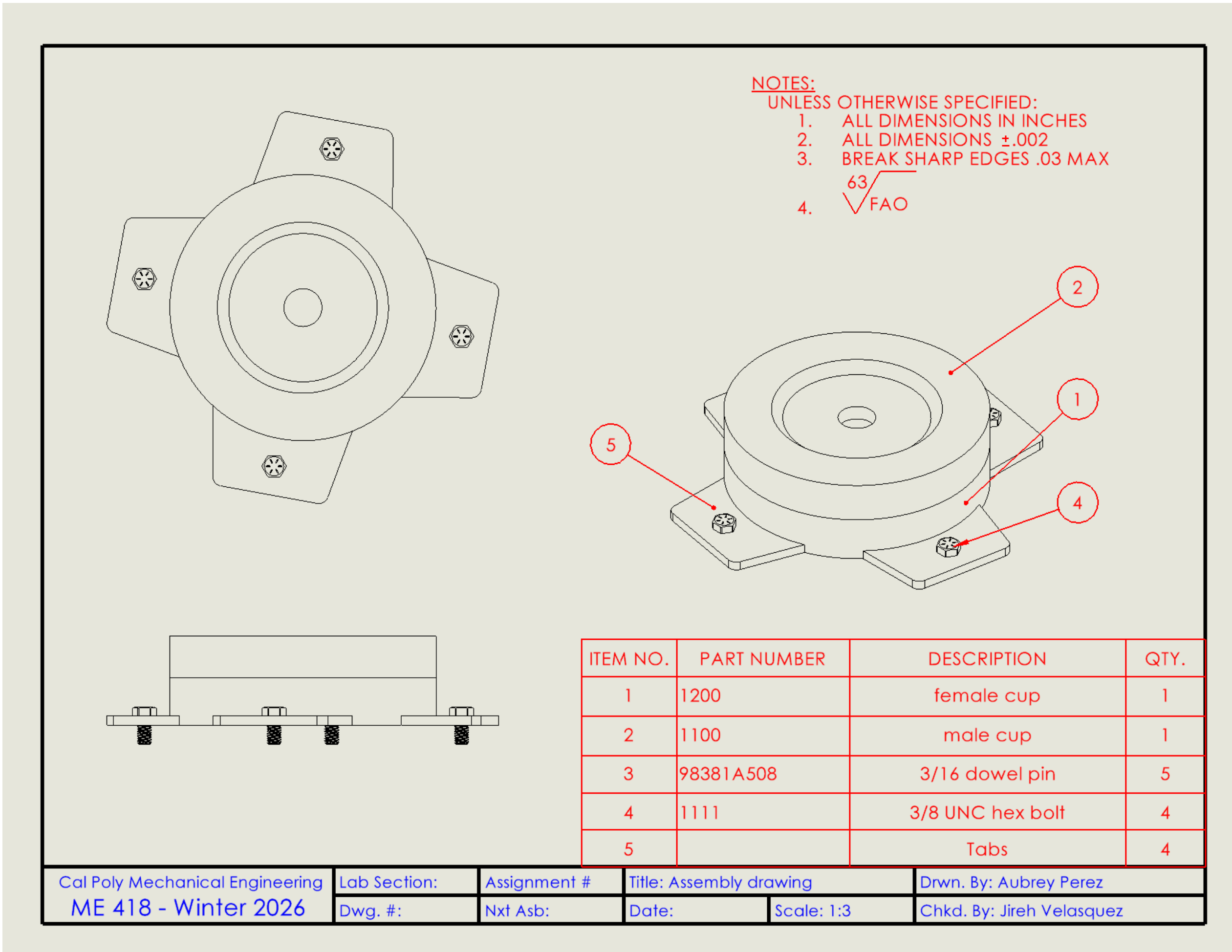


Figure E-4: Full assembly drawing for assemblies 1000 and 2000

Assembly Drawing – Spec Sheet

Item	Specification
Title	Assembly drawing
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:3
Course	ME 418 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	1200	female cup	1
2	1100	male cup	1
3	98381A508	3/16 dowel pin	5
4	1111	3/8 UNC hex bolt	4
5		Tabs	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Verify cup outer lip OD/step diameters and step heights per part drawings before assembly

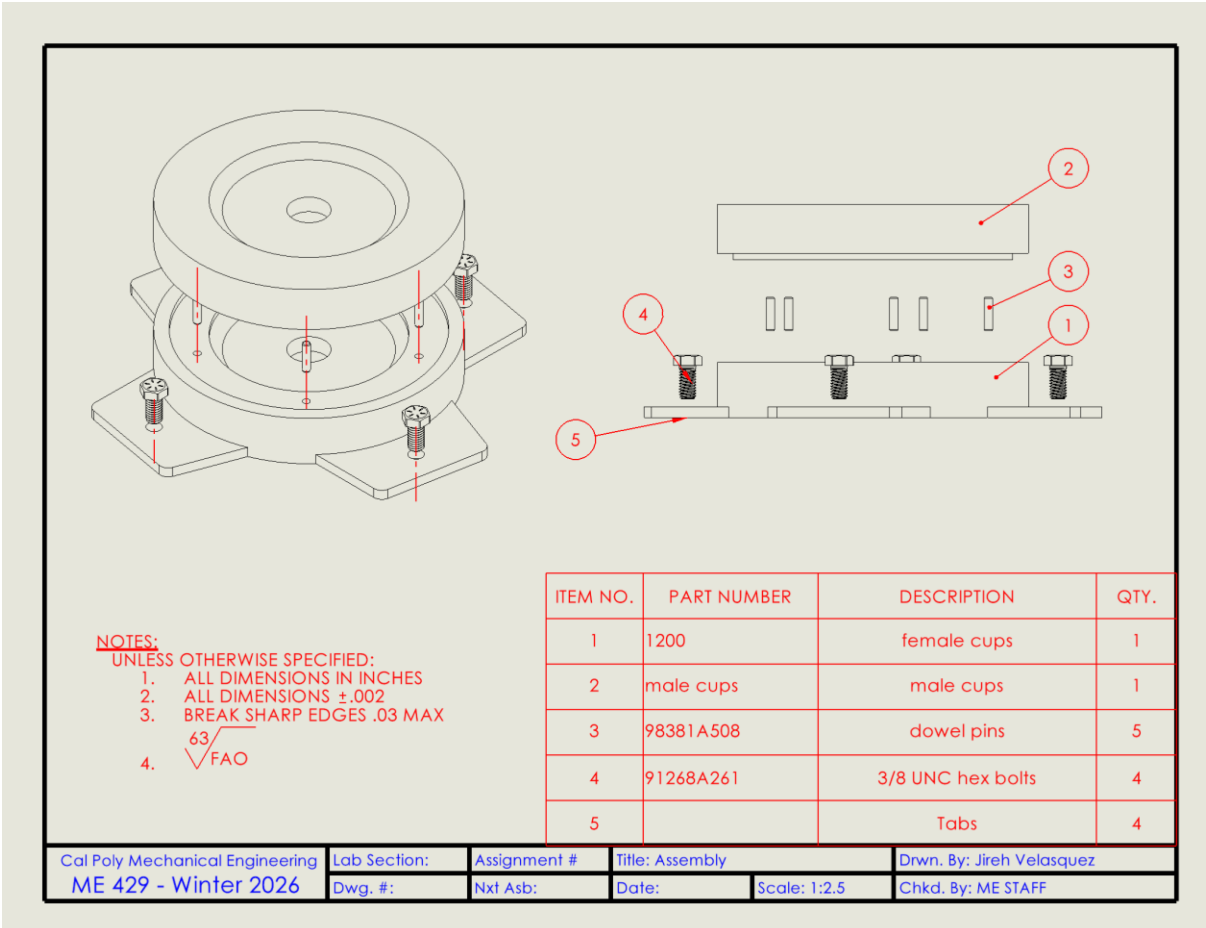


Figure E-5: Exploded assembly view for assemblies 1000 and 2000

Assembly (Exploded) – Spec Sheet

Item	Specification
Title	Assembly
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Jireh Velasquez (sheet shown)
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2.5
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	1200	female cups	1
2		male cups	1
3	98381A508	dowel pins	5
4	91268A261	3/8 UNC hex bolts	4
5		Tabs	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Same as assembly: fit depends on cup lip OD/step diameters and step heights from detail part prints

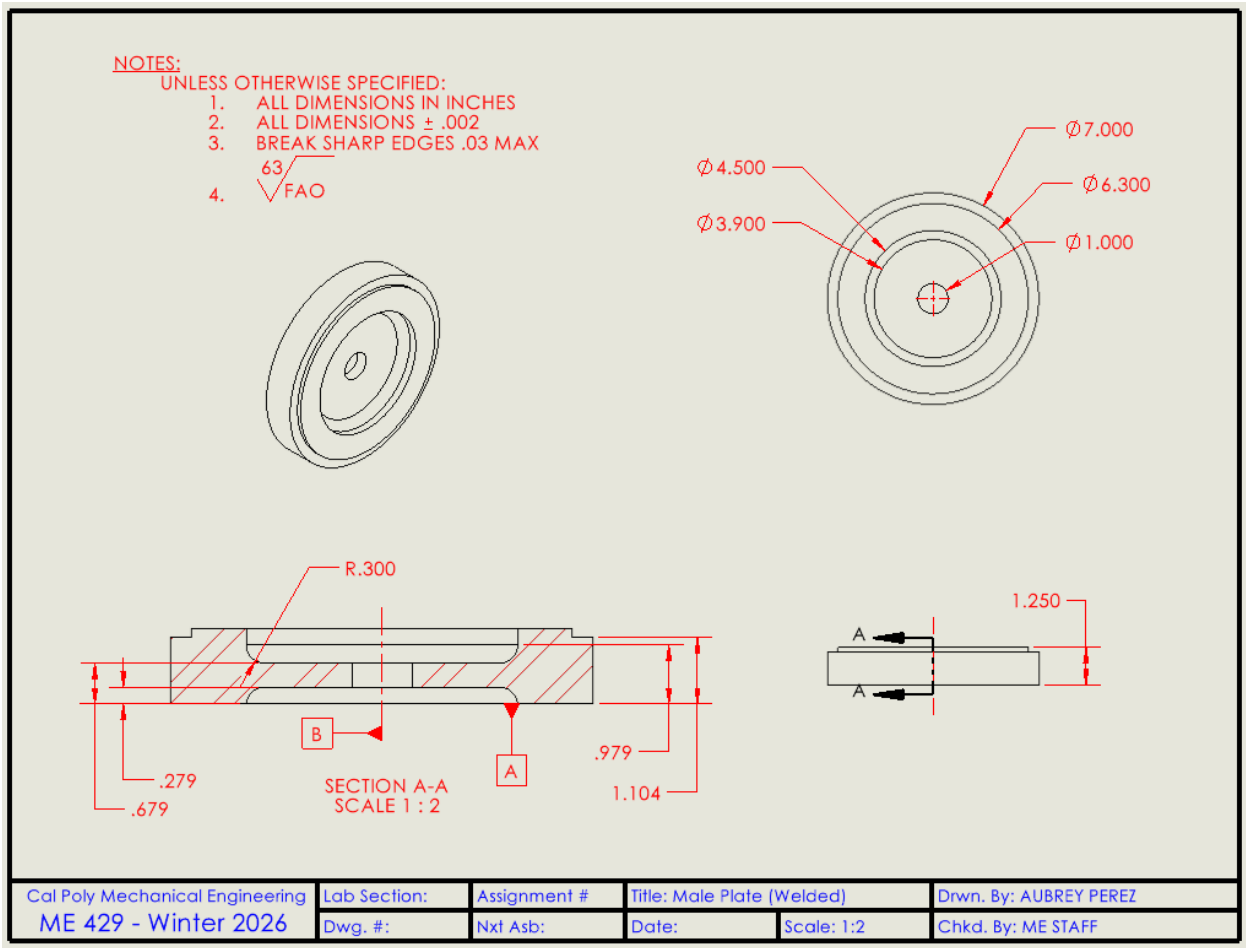


Figure E-6: Detailed drawing of Male Plate for assembly 3000

Male Plate (Welded) – Spec Sheet

Item	Specification
Title	Male Plate (Welded)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Concentric lip/step diameters	Ø6.300, Ø4.500, Ø3.900
Lip fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104

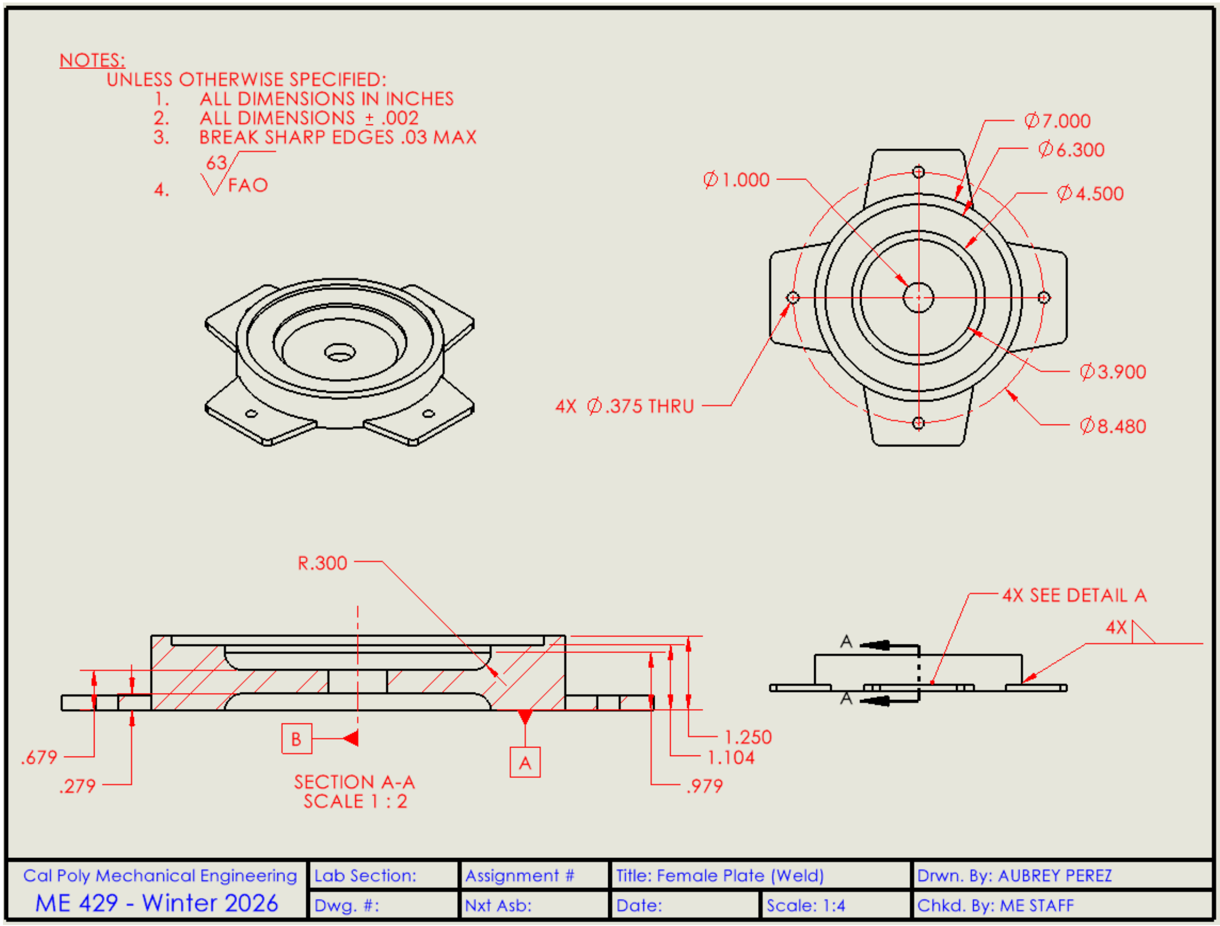


Figure E-7: Detailed drawing of Female Plate for assembly 3000

Female Plate (Weld) – Spec Sheet

Item	Specification
Title	Female Plate (Weld)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:4
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Concentric lip/step diameters	Ø6.300, Ø4.500, Ø3.900
Tab outside diameter/extent	Ø8.480
Lip fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000
Thru holes	4X Ø0.375 THRU

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104, 1.250

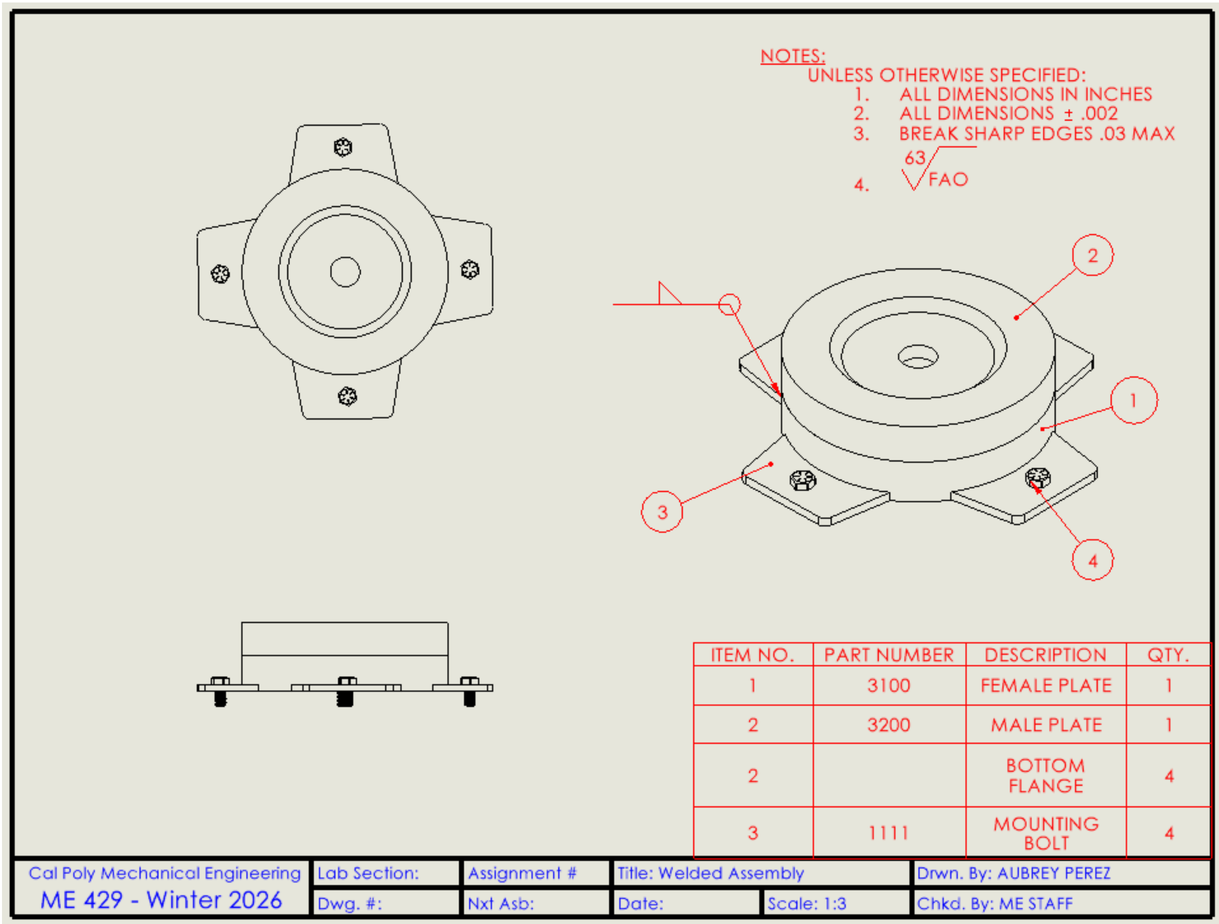


Figure E-8: Detailed welded assembly drawing for assembly 3000

Welded Assembly – Spec Sheet

Item	Specification
Title	Welded Assembly
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:3
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	3100	FEMALE PLATE	1
2	3200	MALE PLATE	1
3		BOTTOM FLANGE	4
4	1111	MOUNTING BOLT	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Maintain plate lip OD/step diameters and lip heights from welded part drawings before weld-up/assembly